Proof of Canopy

Afforestation funding, released only while the forest is still standing.

Live site proofofcanopy.zichengyu.com (reads Base Sepolia live)
Code github.com/zicheng-yu/proof-of-canopy
Video proofofcanopy.zichengyu.com/video (3:26, recorded live on Base Sepolia)

3 plots
real, 250 ha each, chosen from data

93 months
of Sentinel-2 L2A per plot, 2019-01 to 2026-09

15 plot-years
settled on Base Sepolia, 61 transactions

41 tests
Foundry, incl. fuzz on the escrow invariants

01 The problem

Forest carbon projects promise decades of storage, but the money and the credits often move once, after a single verification near planting. If the trees burn or are cleared in year two, nothing in the payment flow reacts, and whoever certified the plot carries no consequence for having been wrong. Buyers end up paying for a promise rather than for forest that is still there.

02 What Proof of Canopy does

Funding sits in an on-chain escrow and is released one year at a time, in proportion to how much of the plot's 2019–2020 canopy Sentinel-2 still sees. At or below a threshold fixed when the project registers (here $T = 70\%$), that year's tranche is frozen; a plot that recovers still earns its later tranches, and whatever stayed frozen returns to the funder after the last year. The survival figure is reported by a verifier who stakes on every claim. Anyone can challenge a claim inside a window by posting a bond, and whoever turns out to be wrong loses their deposit: half to the other side, half burned.

We measure canopy, not carbon. Tonnes of CO_2 per hectare stay with each project's own methodology. This is a monitoring, reporting and verification (MRV) and payout layer: it adds year-by-year proof that the trees behind those tonnes are still there.

03 Five years, three real plots

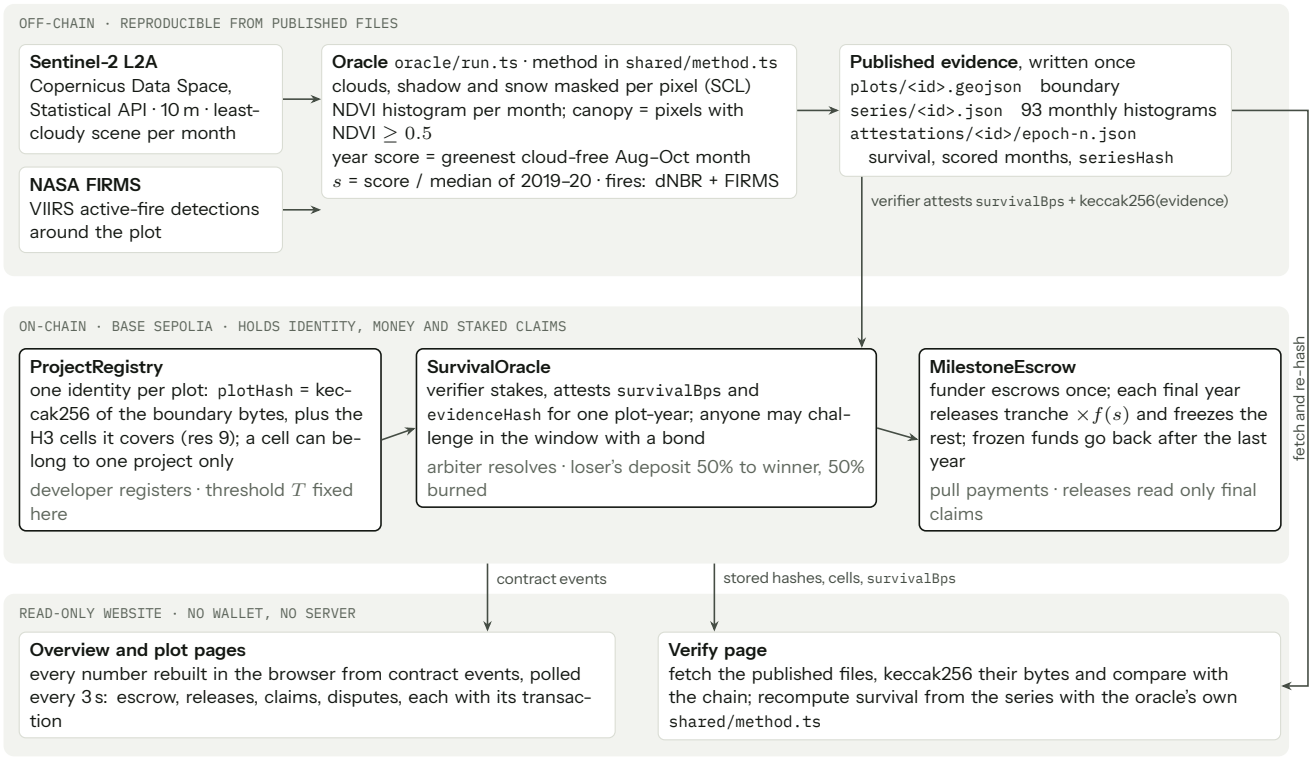
Plot	2021 (E0)	2022 (E1)	2023 (E2)	2024 (E3)	2025 (E4)	Released, ETH
Yedigöller healthy	100% → 100%	100% → 100%	100% → 100%	100% → 100%	99.8% → 99.4%	0.002995
Novo Progresso degraded	100% → 100%	81.7% → 38.9%	86.9% → 56.4%	86.1% → 53.8%	89.4% → 64.6%	0.001798
Manavgat burned	16.2% → 0%	14.5% → 0%	26.6% → 0%	47.5% → 0%	50.9% → 0%	0.000000

Each cell: canopy retained against the 2019–2020 baseline → share of that year's tranche released, as settled on chain. Tranches are 10/15/20/25/30% of 0.003 ETH per project (testnet amounts). ● holding ($\geq 95\%$) ● degrading ● at or below T, tranche frozen. After the last epoch the funder reclaimed the 0.004207 ETH that stayed frozen.

04 What is built and running

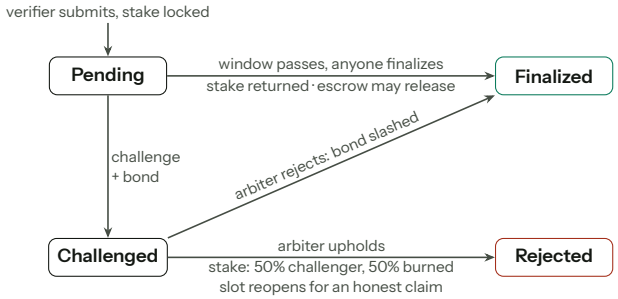
- **Satellite evidence pipeline** (`oracle/`): picks plots from a grid scan, fetches monthly Sentinel-2 statistics from the Copernicus Data Space and NASA FIRMS fire detections, and writes one evidence file per plot-year that cites the monthly series by hash.
- **Three Solidity contracts on Base Sepolia**, sources verified on Sourcify: `ProjectRegistry`, `SurvivalOracle`, `MilestoneEscrow`. The full story ran on them: registration, a refused double registration, funding, all 15 plot-years settled, a staged false claim challenged and slashed, frozen money reclaimed and earned money withdrawn.
- **A read-only website** that rebuilds every number from contract events and lets anyone re-hash the evidence and recompute survival in the browser, with the same code the oracle runs.
- **Escrow invariants under fuzzing:** tranches always sum to the escrowed amount, a higher survival never releases less, and every wei of a settled tranche is either released or frozen.

05 Architecture

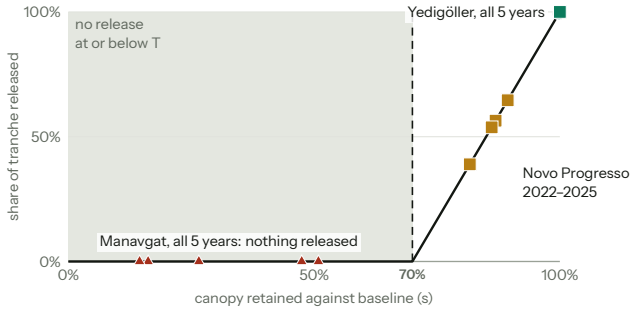


The measurement stays off chain and can be rerun by anyone from the published files; the chain holds what no single party should hold (the plot's identity, the money, and a public, staked record of who claimed what); the website only reads.

Life of one claim



Release rule, with all 15 settled plot-years



$f(s) = \text{clamp}\big((s - T)/(1 - T),\ 0,\ 1\big), T = 70\%.$

● Yedigöller ■ Novo Progresso ▲ Manavgat

06 Verify it yourself

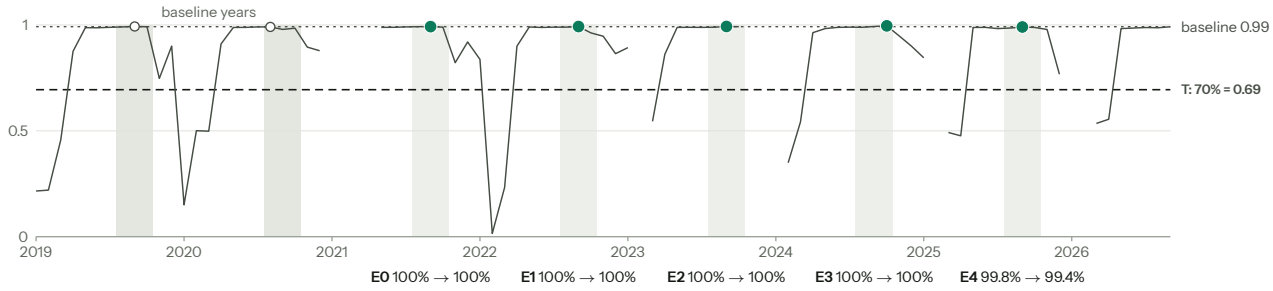
The website's Verify page runs four checks in the browser for any attestation. For the staged false claim (attestation #15, "Manavgat recovered to 94% in 2022"), the file is genuine but the number is not:

Check	What the browser does	Attestation #15
Evidence file matches the claim	keccak256 of the published file's bytes vs. `evidenceHash` on chain	● match
Boundary matches the plot	keccak256 of the GeoJSON bytes vs. `plotHash`	● match
Cells match the boundary	recompute the H3 res-9 cells with h3-js vs. `cellsOf`	● 22 of 22 identical
Survival follows from the satellite series	check `seriesHash`, rerun `shared/method.ts` on the series vs. `survivalBps`	● recomputed 14.49%, claimed 94%: mismatch

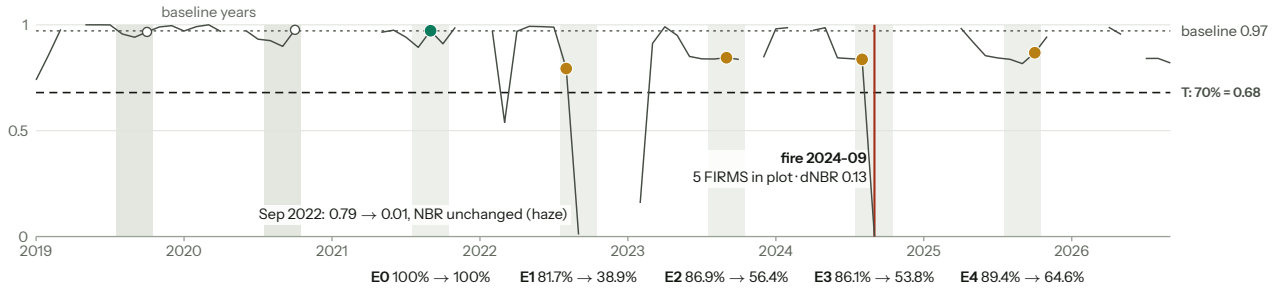
07 Measuring canopy

1. **Pick plots from data, not by eye.** `oracle/scan.ts` scores a grid of 250 ha squares by late-season canopy in a baseline and a test year. For the burned plot the grid is centred on the densest cluster of FIRMS detections during the July–August 2021 Manavgat fire.
2. **Fetch the series.** One Statistical API request per plot returns 93 months (2019-01 to 2026-09) of Sentinel-2 L2A at 10 m: the least cloudy scene of each month, clouds, shadow, cirrus and snow masked per pixel, and a 40-bin NDVI histogram. Canopy fraction = share of cloud-free pixels with NDVI ≥ 0.5; months under 60% cloud-free are not used.
3. **Score each year on its greenest Aug–Oct month**, against the median of 2019 and 2020. Clouds, haze and smoke only ever lower NDVI, so the maximum is the reading least contaminated by them, while real canopy loss lowers every month. Survival s is the ratio, capped at 1, in basis points.
4. **Write evidence once.** Each plot-year file carries the baseline, the observed value, the months used, s, and `seriesHash` = keccak256 of the series file. The chain stores keccak256 of the evidence file itself.

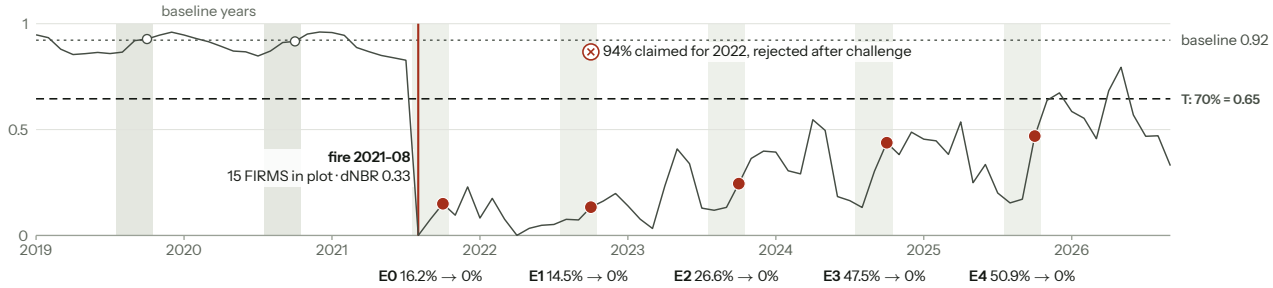
Yedigöller · Bolu, Türkiye · national park forest



Novo Progresso · Pará, Brazil · deforestation arc



Manavgat · Antalya, Türkiye · burned in the July–August 2021 wildfire



Monthly canopy fraction of each 250 ha plot (line; gaps are cloudy months). Shaded: the scored Aug–Oct seasons, darker for the baseline years. Dots: the month each year was scored on, coloured by outcome; hollow dots: baseline months. At Yedigöller the winter dips are leaf-off broadleaf trees and low sun; that and clouds are why only Aug–Oct is scored. Under each year: canopy retained → share of that tranche released.

What the series show. The Manavgat fire is confirmed three independent ways: canopy falls from 0.83 to 0.00 in 2021-08, the burn ratio drops by dNBR 0.33, and 15 NASA FIRMS detections fall inside the plot on 2021-08-01 and 2021-08-02. Over the Amazon plot, a September 2022 reading falls from 0.79 to 0.01 while NBR holds at 0.54 and the canopy is back the next season: haze, which is why each year is scored on its greenest month rather than its mean. The September 2024 drop there does carry a burn signature (dNBR 0.13, 5 FIRMS detections in plot) and is recorded in that year's evidence.

08 Why this needs a chain

Why not a database?	Its owner is always one of the three parties: funder, developer or verifier. Here nobody has to own the ledger, and a plot's identity is unique across everyone who uses it.
Why not an escrow agent?	Escrow can lock money, but not the number that unlocks it: whoever computes survival controls the payment. Here the computation is reproducible from hashed evidence and the reporter is staked.
Why not just signatures?	A signature shows who said something. It does not stop them from saying two different things, and lying costs nothing. One live claim per plot-year, a stake behind it and a public challenge window make a false claim visible and expensive.
Is the measurement on chain?	No, and we do not pretend it is. The chain holds the economic game and the evidence hashes; anyone can rerun the method on the same series and get the same number, as the Verify page does.

09 Who uses it, and how it gets adopted

Users

- **Funders and credit buyers** (corporate pre-purchase, climate funds) paying for outcomes, not promises.
- **Project developers** raising staged money against a credible, public survival record.
- **Independent verifiers** who earn by staking on accurate claims and lose stake on wrong ones.
- **Registries** that need a guard against the same land being registered twice.

Adoption path

1. Run beside an existing project as a monitoring layer; its carbon methodology is untouched.
2. A funder escrows one pilot tranche against the project's real boundary.
3. Open the verifier role to staked third parties; challengers keep them honest.
4. Replace the single arbiter with a multisig or dispute protocol; add radar or lidar so canopy *structure*, not only greenness, is measured.

10 Cost and scale

Action (Base Sepolia run)	Gas
register a plot (21–22 H3 cells)	783,214
fund the escrow	78,658
stake	44,998
attest one plot-year	179,988
finalize	41,837
release one tranche	100,256
challenge / resolve	118,602 / 68,150
settle one plot-year (attest + finalize + release)	**322,081**

- **Data:** one API request returns a plot's whole 93-month history; three plots take about 40 s on the free Copernicus tier.
- **Chain:** a plot costs one registration and three cheap calls a year; registration grows with the plot's cell count (one storage slot per H3 cell).
- **Double counting:** H3 is a global grid, so "has this land already been claimed?" is a lookup, for any registry that reads the contract.

11 Limits, stated plainly

- Disputes are decided by one arbiter address in this version; production needs a multisig or dispute protocol.
- NDVI sees the canopy surface. Thinning under a closed canopy barely moves it, and shrub regrowth counts once its NDVI passes 0.5: part of Manavgat's climb to 50.9% by 2025 is maquis, not pine. The 70% threshold keeps that from releasing money.
- Scoring on the greenest month means a loss after it is counted the following year (the September 2024 burn at Novo Progresso falls after that year's August score).
- Plots are 250 ha squares placed by our scan, standing in for real project boundaries.

12 Deployment, reproduction and sources

Base Sepolia (chain 84532), sources verified on Sourcify

`ProjectRegistry`	`0x9b63…da2d`
`SurvivalOracle`	`0x5BA5…4702`
`MilestoneEscrow`	`0xb2AC…4a39`

Every transaction of the run: `deployments/84532.demo.json` and each plot's page on the site.

Reproduce

`npm test`	41 Foundry tests
`npm run oracle`	Sentinel-2 + FIRMS → evidence
`npm run demo:local`	whole story on a local chain
`npm run demo:testnet`	deploy and run on Base Sepolia

Contains modified Copernicus Sentinel data (2019–2026) via the Copernicus Data Space Ecosystem. Fires: NASA FIRMS, VIIRS S-NPP. Site basemap: Sentinel-2 cloudless by EOX. Fonts: Instrument Sans, IBM Plex Mono (SIL OFL).